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AQA

Please write clearly in block capitals.

Centre number | | | | | | Candidate number | | | | |

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I declare this is my own work.

Functional Skills Level 2 MATHEMATICS

Paper 2 Calculator

Monday 9 January 2023 Afternoon Time allowed: 1 hour 30 minutes

Materials

For this paper you must have:
- a calculator
- mathematical instruments.



Instructions
- Use black ink or black ball-point pen. Draw diagrams in pencil.
- Answer **all** questions.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do all rough work in this book. Cross through any work you do not want to be marked.

For Examiner's Use	
Question	Mark
1–6	
7	
8	
9	
10	
TOTAL	

OFFICIAL MFS MODEL SOLUTION · BENCHMARK EXEMPLARY PASS

Full Marks / Exemplary (100% Pass · Model Solutions)

· Assessment: January 2023 - Paper 2

· Subject: AQA Functional Skills Maths Level 2

· Complete accredited step-by-step models and tutor guidance

· Exam Board: AQA (Ofqual Regulated Reform Standard)

Verified by Tutor Abdul (Senior Assessor) · MasterFunctionalSkills.co.uk

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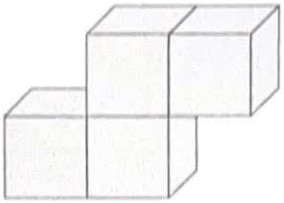
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Section A

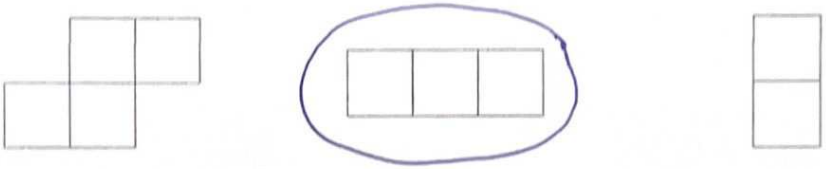
Answer **all** questions in the spaces provided.

1 Here is a solid made from four cubes.



Circle the plan view of the solid.

[1 mark]



2 Write the number 2 307 049 in words.

[1 mark]

Two million, three hundred and seven thousand and forty nine.

Do not write outside the box

3 Which is closer in value to 0.5

0.379 or 0.614 ?

You **must** show your working.

[2 marks]

$0.614 - 0.5 = 0.114$

$0.5 - 0.379 = 0.121$

$0.114 < 0.121$ so 0.614 is closer in value to 0.5

Answer 0.614

4

Surface area of a sphere = $4\pi r^2$

Work out the surface area of a sphere with a radius of 2.3 cm
State the units of your answer.

[3 marks]

$4 \times \pi \times 2.3^2 = 4 \times \pi \times 2.3 \times 2.3$

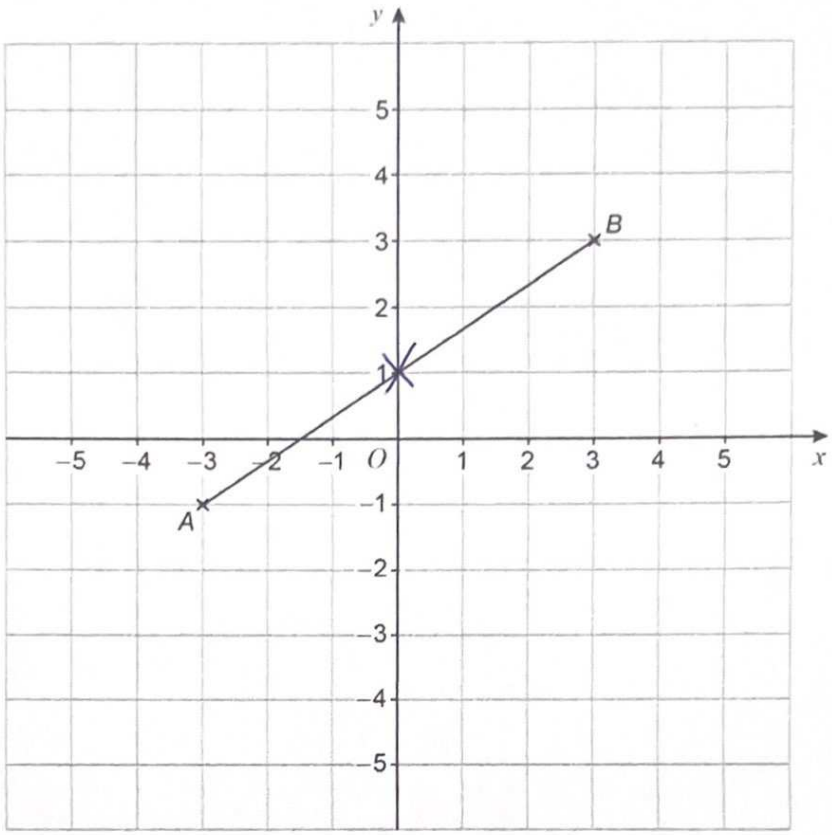
$= 21.16\pi$

$= 66.48$

Answer 21.16π or $66.48\ cm^2$

Do not write outside the box

5



Point *C* is the midpoint of line *AB*.

Plot point *C* on the grid and write down the coordinates of *C*.

[2 marks]

see graph

Answer (0 , 1)

Do not write outside the box

6 The table contains grouped data.

x	Frequency	Mid-point	
$0 < x \leqslant 20$	7	10	70
$20 < x \leqslant 40$	9	30	270
$40 < x \leqslant 50$	4	45	180
	Total = 20		520

Work out an estimate of the mean of x.

[3 marks]

See table

7 x 10 = 70

9 x 30 = 270

4 x 45 = 180

70 + 270 + 180 = 520

520 ÷ 20 = 26

Answer 26

12

Turn over for Section B

Do not write outside the box

Section B

Answer **all** questions in the spaces provided.

7 School office

Ashley works in a school office.

7 (a) Ashley needs 29 book bags.

He sees these offers online.

Best book bags	**Bags and stuff**
Book bags £4.99 each For every 7 you buy you get an extra 1 free Free delivery	Book bags reduced to £4.25 each Delivery charge £3.90 per order

Ashley wants the cheapest total price for the 29 bags and delivery.

Which offer should he choose?
You **must** show your working.

[5 marks]

Best Book Bags

29 ÷ (7+1) = 3.625 (3 free)

29 - 3 = 26 26 x 4.99 = 129.74

Bags and stuff

29 x 4.25 + 3.9 = 127.15

127.15 < 129.74

so should choose Bags and Stuff.

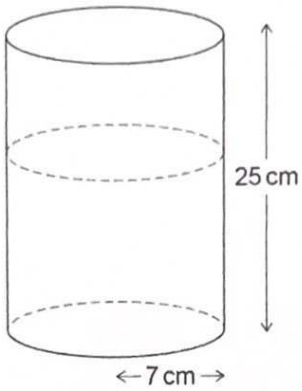
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7 (b) Ashley needs to put drinking water into 40 jugs before break time.

Each jug

is a cylinder with radius 7 cm and height 25 cm

will be $\frac{3}{5}$ full.



Water flows out of the tap at 13 litres per minute.

$1000\,cm^3 = 1$ litre

Can Ashley put the right amount of water into all 40 jugs in 6 minutes?
You **must** show your working.

[6 marks]

Volume $= \pi r^2 L = \pi \times 7 \times 7 \times 25$

$= 1225\pi$

$3/5 \times 1225\pi = 735\pi = 2309.1$

$40 \times 2309.1 \div 1000 = 92.36...$ how many litres she needs to fill

$13 \times 6 = 78$ Litres per minute $< 92.36...$

No she does not have time

11

Do not write outside the box

8 Lifeguard

Talia is a lifeguard at a swimming pool.

8 (a) One morning, Talia checked the water temperature in the pool each hour.

Time	8 am	9 am	10 am	11 am
Temperature in °F	75	80	81	87

The formula to convert Fahrenheit (F) into Celsius (C) is

$$C = \frac{5}{9}(F - 32)$$

What time was the first check where the water temperature was above 26°C?
You **must** show your working.

[3 marks]

8am $C = \frac{5}{9}(75 - 32) = \frac{5}{9}(43) = 23.888\ldots < 26$

9am $C = \frac{5}{9}(80 - 32) = \frac{5}{9}(48) = 26.66\ldots > 26$

Answer 9am

Do not write outside the box

8 (b) There are 90 people in the pool.

number of adults : number of children = 3 : 2

One lifeguard is needed for every 10 children.

How many lifeguards are needed for the children in the pool?

[4 marks]

$90 \div (3+2) = 18$

$18 \times 2 = 36$ children present

$36 \div 10 = 3.6$

so 4 lifeguards are needed

Answer 4

7

Turn over for the next question

Do not write outside the box

9 Hedgehogs

Aahad works for a charity that rescues hedgehogs.

9 (a) The charity has four main costs each month.

	Cost per month
Straw	£25
Feed	£39
Electricity	£68
Vets	£235

From next month the costs are changing.

- Straw will be free from a local farm
- Feed is increasing by $\frac{1}{6}$
- Electricity is increasing by £324 a year
- The cost of the vets will be £3090 a year.

The charity receives a new monthly donation of £30

Is the new donation enough to pay for the **increase** in total monthly costs?
You **must** show your working.

[5 marks]

feed 39 x 1/6 = 6.5 increase

electricity 324 ÷ 12 = 27 increase

vets 3090 ÷ 12 − 235 = 22.5 increase

straw 25 decrease

6.5 + 27 + 22.5 − 25 = 31 > 30

no the extra donation won't cover the increase in ~~bills~~. monthly costs.

Do not write outside the box

9 (b) The charity needs a new building for the hedgehogs.

The new building will cost £250 000

The charity invested £245 000 in a bank account 2 years ago.

The account pays 1.2% compound interest each year.

Is there enough money in the bank account to pay for the building?
You **must** show your working.

[3 marks]

1.2% 1.012 multiplier

$245000 \times 1.012^2 = 250\,915.28$

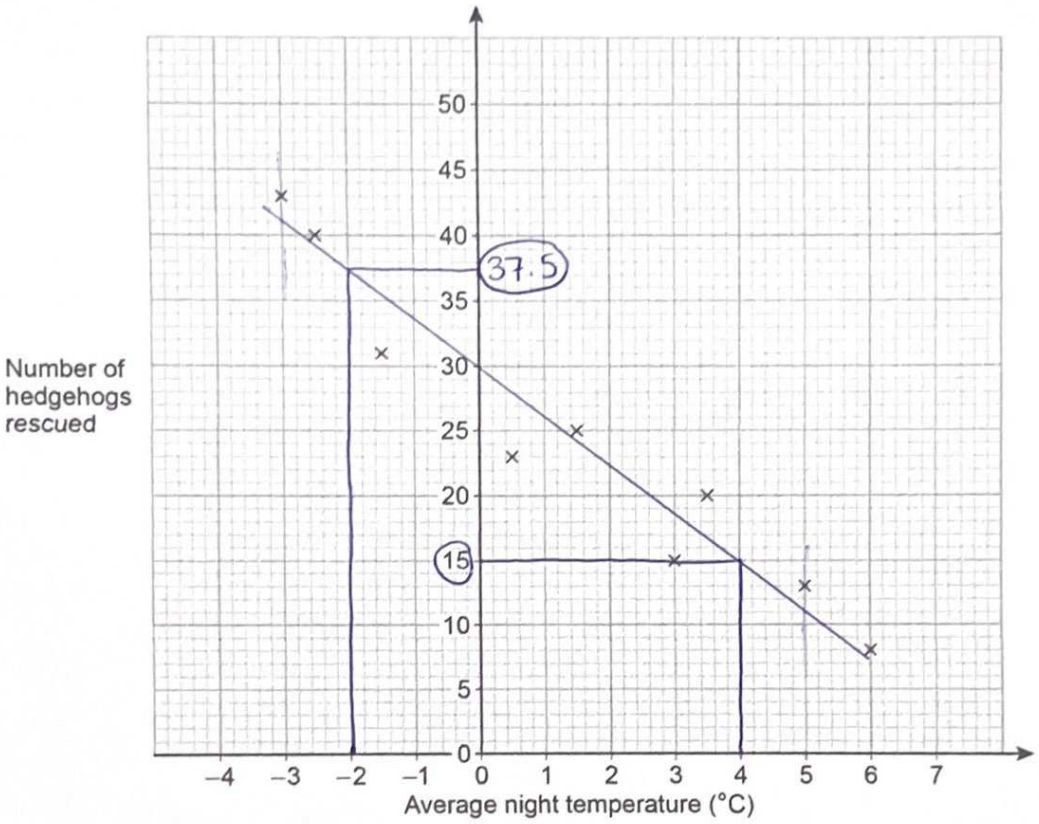
so yes there is enough money in the account to pay for the building.

Question 9 continues on the next page

Do not write outside the box

Each week Aahad records the number of hedgehogs rescued and the average night temperature that week.

The scatter diagram shows information for the last 9 weeks.



9 (c) Describe the relationship between average night temperature and number of hedgehogs rescued.

[1 mark]

As temperature increases the number of hedgehogs rescued decreases

Do not write outside the box

9 (d) In week A the average night temperature was –2°C

In week B the average night temperature was 4°C

Use the scatter diagram to estimate how many **more** hedgehogs were rescued in week A than in week B.

You **must** show your working, some of which should be on the diagram.

[4 marks]

-2°C 37.5 on average

4°C 15 on average

37.5 - 15 = 22.5

dependent on your line of best fit.

Answer 22.5

13

Turn over for the next question

Do not write outside the box

10 **Housework**

David is doing housework.

10 (a) David is washing clothes.





David says,

"I can do **at least** 32% more washes with the powder than the liquid."

Is he correct?
You **must** show your working.

[5 marks]

WP: $3900 \div 65 = 60$ washes

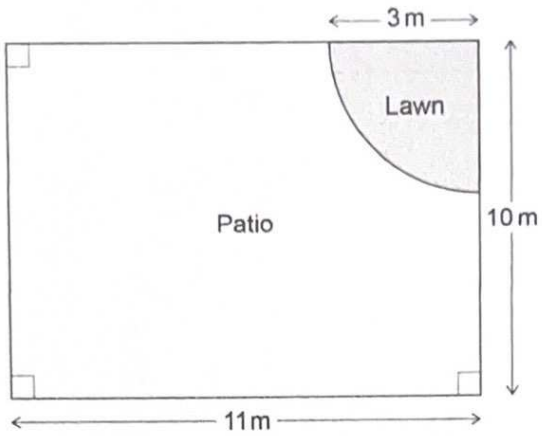
WL: $1540 \div 35 = 44$ washes

$\frac{60 - 44}{44} \times 100 = 36\ \%$

$36\ \% > 32\ \%$ so David is right.

Do not write outside the box

10 (b) David is going to clean the patio.



Not drawn accurately

The lawn is a quarter circle with radius 3 m

Each bottle of cleaner

cleans 15 m^2 of patio

costs £1.95

Work out the cost of the bottles of cleaner that David needs to clean the patio.
You **must** show your working.

[6 marks]

$11 \times 10 = 110\ m^2$ total area

$\frac{1}{4} \pi \times 3^2 = \frac{9}{4} \pi\ m^2$ area of a lawn

$110 - \frac{9}{4} \pi = 102.93\ m^2$ area of patio

$102.93 \div 15 = 6.86$ need 7 bottles

$7 \times 1.95 = 13.65$

Answer £ 13.65

Question 10 continues on the next page

Do not write outside the box

10 (c) David makes a cleaning spray using vinegar and water.

The cleaning spray needs to be

700 ml in total

45% vinegar.

David has put **half a pint** of vinegar in the spray bottle.

How much **more** vinegar and how much water does he need to add?
Use 1 pint = 568 ml
State the units of your answers.

[6 marks]

0.45 × 700 = 315 ml of vinegar

0.5 × 568 = 284 ml of vinegar already in

315 − 284 = 31 ml more vinegar

700 − 315 = 385 ml of water

Vinegar 31

Water 385

17

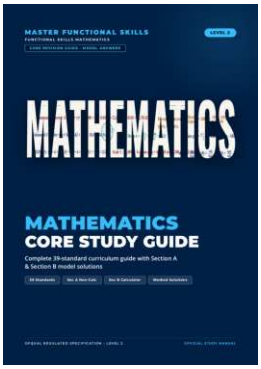
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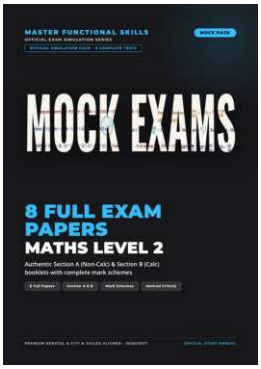


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